

ASHUTOSH RAVAL

Senior Node.js / Full-Stack Backend Developer · 7+ Years

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PROFESSIONAL SUMMARY

Senior Node.js engineer with 7 years at Webelight Solution, deliberately deepening expertise across 8+ independent client verticals — logistics, fintech, insurance, real estate, and SaaS — each with its own architecture, domain, and scale challenges. Designed a real-time chat microservice sustaining 100k+ concurrent users at sub-100ms p99 latency, and a multi-tenant booking platform that eliminated double-bookings across 500+ daily reservations using optimistic concurrency control. Promoted to Senior in 2020; since then led architecture for 8+ microservice products and mentored junior engineers — 2 of whom reached mid-level within 18 months. Stayed because the technical growth was real; now actively seeking a new environment where I can bring that depth to a larger product and a broader team.

PROFESSIONAL EXPERIENCE

Senior Node.js Developer · Webelight Solution Pvt. Ltd., Ahmedabad, India *January 2020 – Present*

Architecture & Technical Leadership

- Owned architecture decisions for 8+ microservice products; patterns adopted as company-wide baseline — reduced onboarding time for new engineers joining existing codebases.
- Defined performance SLAs for all high-traffic systems and met or exceeded them across every production deployment; zero SLA breaches in 5 years as senior engineer.
- Mentored a rotating cohort of 4–6 junior engineers per sprint; 2 progressed to mid-level within 18 months through structured code review and pair programming.
- Led sprint planning and task allocation for cross-functional teams of up to 8 engineers across 3 concurrent product tracks.

Backend Engineering

- Built event-driven microservices with Kafka and Socket.IO, enabling independent deployments and decoupled service boundaries.
- Delivered REST and GraphQL APIs (Apollo, TypeGraphQL, Prisma) with consistent sub-200ms p95 latency under production load across multiple client apps.
- Reduced average query latency by ~45% on the most-accessed endpoints through targeted MongoDB aggregation pipeline rewrites and Redis caching layer design.
- Achieved 95% test coverage across critical payment and booking paths using Jest; identified and fixed 3 race-condition bugs pre-launch that would have caused data loss.
- Enforced server-side RBAC across multi-role platforms (5+ stakeholder types), ensuring role permissions are validated at the API layer and cannot be bypassed via frontend routing.

Cloud & DevOps

- Deployed serverless workloads on AWS Lambda; containerised services with Docker Compose to eliminate environment-specific bugs across dev and production.
- Integrated Google Cloud Functions, Firebase Auth, and Firestore for real-time auth flows across 3 products, reducing auth-related support tickets.
- Automated CI/CD pipelines; implemented CloudWatch alerting with custom health-check endpoints — cut mean time to detect (MTTD) for production issues.

Node.js Developer (Mid-level) · Webelight Solution Pvt. Ltd., Ahmedabad, India *July 2018 – December 2019*

- Built and shipped full-stack features across 3 client products under senior guidance; took ownership of API design and database schema from day one.
- Introduced Redis caching on a high-read endpoint, cutting p99 response time from ~800ms to ~120ms — first major performance win recognised by the team.

- Wrote initial Jest test suites for a greenfield fintech module; practice later adopted team-wide as the standard testing approach.

KEY PROJECT HIGHLIGHTS

Real-Time Chat Microservice

Stack: Node.js · MongoDB · Redis Pub/Sub · Socket.IO · GraphQL · Docker · AWS

- Engineered horizontal scaling with load balancing across multiple Node.js instances, sustaining 100k+ concurrent users with sub-100ms p99 message delivery latency under peak load.
- Implemented Redis Pub/Sub for distributed presence tracking and cross-instance notification broadcast, eliminating missed messages during instance failover.
- Built GraphQL subscription APIs for live data sync, removing the need for client-side REST polling entirely — reduced unnecessary API calls by an estimated 70%+.
- 95% test coverage on critical paths; zero P0 production incidents in the first 6 months post-launch across a userbase that scaled from 0 to 100k concurrent sessions.

Multi-Tenant Slot Booking Platform (B2B SaaS)

Stack: Node.js · Express.js · MongoDB · Razorpay · WebSockets

- Designed row-level data isolation for ~15 independent client organisations from a single deployment — no data cross-contamination incidents across 12 months in production.
- Implemented optimistic concurrency control that eliminated double-bookings: 0 conflicting reservations recorded across ~500 daily bookings in production.
- Built dynamic slot allocation engine with real-time calendar sync, cutting manual scheduling overhead for client operations teams by an estimated 3+ hours/day.
- Integrated Razorpay for full booking-to-settlement payment flows including automated payouts — removed the need for manual reconciliation on every transaction.

CCM — Construction & Contractor Management (Real Estate SaaS)

Stack: Node.js · Express · PostgreSQL · Sequelize · React · JWT · RBAC

- Architected an end-to-end real estate project lifecycle platform serving 5 distinct stakeholder roles (Super Admin, Admin, Inspector, Contractor, Client) from a single deployment.
- Designed a linked financial pipeline (Quotation → Job → Invoice) with PostgreSQL foreign-key referential integrity, ensuring no billing step could be skipped or corrupted.
- Enforced server-side RBAC at the API layer — role permissions validated before any data access — eliminating the risk of frontend-bypass and cross-role data leakage.
- Built role-scoped dashboard aggregations (weekly/monthly/yearly) computed server-side, minimising over-fetching and delivering contextual analytics per stakeholder type.
- Implemented a scheduling module as a first-class entity linking reminders to projects, activities, and target roles — reducing missed deadlines across multi-stakeholder workflows.

Insurance Bar API — Multi-Role Insurance Management Platform

Stack: Node.js · Express · MongoDB · PostgreSQL · JWT · RBAC · React

- Built a comprehensive insurance lifecycle platform with 5 roles (Super Admin, Admin, Customer, Partner, Partner Associate), live at admin.theinsurancebar.com.
- Delivered end-to-end policy management engine covering creation, updates, partner attachment, lifecycle transitions, and expiration — all with server-side role gating.
- Engineered a multi-stage claims and case management workflow with configurable approval/rejection mechanisms and status tracking across the full resolution lifecycle.
- Integrated loan management with approval workflow, subscription-based benefit allocation for partner networks, and payment processing for fees and subscriptions.
- Built a support ticketing system with status tracking, ensuring full auditability of user issues from creation through to resolution.

Freight Booking & Logistics Platform — Navodaya Transport

Stack: Node.js · Express.js · MongoDB · Razorpay · Traqo GPS

- Built role-based access system (Loader, Trucker, Admin) managing the end-to-end shipment lifecycle from booking through delivery and payment settlement.

- Integrated Traqo SIM-based GPS API for real-time location tracking, giving dispatchers live visibility of all active shipments without manual check-ins.
- Developed admin dashboard covering user management, booking oversight, and payment reconciliation; reduced dispatcher workload for status reporting by ~2 hours/day.
- Automated booking and status notification system, measurably reducing inbound support queries related to shipment visibility.

Additional Products Delivered

- Order Management System — multi-vendor inventory sync, fulfillment workflows, and automated payment reconciliation for an e-commerce client.

TECHNICAL SKILLS

Runtime & Frameworks	Node.js (Express, Fastify, NestJS), TypeScript, JavaScript (ES6+), microservices, event-driven architecture
Databases	MongoDB (aggregation pipelines), PostgreSQL, MySQL, Firebase Firestore, Realtime Database
Caching & Messaging	Redis (caching, Pub/Sub, rate limiting, sessions), Apache Kafka / KafkaJS, message queues
APIs	REST, GraphQL (Apollo Server, TypeGraphQL), Prisma ORM, Sequelize ORM, WebSockets, Socket.IO
Cloud & DevOps	AWS (Lambda, ECS, EC2, S3, CloudWatch, IAM), GCP (Cloud Functions, Pub/Sub, Firestore), Docker, Docker Compose, Kubernetes, Terraform, CI/CD (GitHub Actions), Monitoring (Prometheus/Grafana)
Auth & Security	OAuth 2.0, Passport.js, JWT, session management, multi-factor authentication, server-side RBAC
Payments	Stripe, Razorpay, Paytm — processing, subscription billing, automated payouts
Tools & CMS	Strapi, Payload CMS, Traqo GPS, Git, NPM/Yarn
AI Coding Tools	Claude (Claude.ai, Claude Code), Cursor, Windsurf, OpenCode, Google Antigravity, Freebuff, GitHub Copilot, OpenAI Codex, Gemini

AI-AUGMENTED DEVELOPMENT

Consistently leverages AI tooling to accelerate the full backend development lifecycle — from architecture planning and code generation through debugging, code review, and documentation. ~85% of day-to-day development output is AI-assisted, with human oversight focused on system design decisions, performance validation, and production quality gates.

Agentic IDEs & Editors

- **Cursor:** Primary AI-native IDE; uses inline chat, codebase-aware completions, and Composer for multi-file edits across all Node.js projects.
- **Windsurf:** Agentic IDE with Cascade flow for autonomous multi-step code tasks; used for refactor-heavy sprints.
- **Google Antigravity:** Google's agent-first platform (powered by Gemini 3 Pro); used for multi-agent orchestration across editor, terminal, and browser to plan, implement, and verify features end-to-end.

Terminal & CLI Agents

- **Claude Code:** Anthropic's terminal agent for agentic coding tasks; used for scaffolding, large refactors, and system design documentation generation.
- **OpenCode:** Open-source TUI agent with free-tier model access; used for rapid prototyping and cost-effective agentic tasks.
- **Freebuff:** Free CLI coding agent (Codebuff platform, ad-supported); multi-agent framework with browser verification — used for zero-friction task delegation without token costs.

- **OpenAI Codex:** Cloud-based agentic coding environment; used for isolated feature branches and background task execution.

AI Assistants & Autocomplete

- **Claude (claude.ai):** Primary AI assistant for architecture discussions, code review, system design, and technical writing.
- **Gemini:** Used for Google ecosystem tasks, research, and as the model backbone in Antigravity workflows.
- **GitHub Copilot:** In-editor autocomplete and chat; used across VS Code for inline suggestions and PR summaries.

EDUCATION

Master of Computer Applications (MCA) · SGPA 8.3/10

2019

Acharya Motibhai Patel Institute — Ganpat University, Gujarat

Bachelor of Computer Applications (BCA)

2015

Hemchandracharya North Gujarat University, Gujarat

ADDITIONAL

- **Publications:** "In-App Server Notification for Android" and "Running Danger.js Locally" on Medium — practical engineering write-ups.
- **Languages:** English, Hindi, Gujarati (professional fluency in all three).
- **Portfolio:** ashup1711.github.io/portfolio — 5 detailed case studies covering real-time systems, SaaS, logistics, real estate, and insurance platforms.
- **Continuous learning:** Actively experimenting with emerging backend patterns; regular reader of Node.js ecosystem release notes and RFCs.